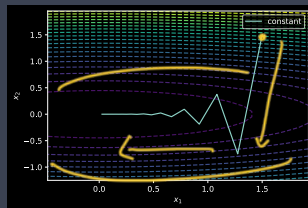
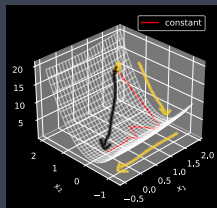


» Momentum Methods: Motivation



- * In steep, narrow valleys moving in direction of steepest descent might not be best. For example:
- * $f(x) = x_1^2 + 10x_2^2$, starting value $x = [1.5, 1.5]$, constant step size $\alpha = 0.15$

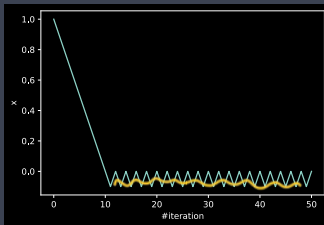
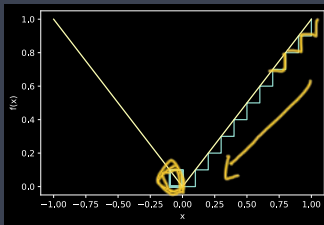


- * Wouldn't it be faster to take a less zig-zag path towards minimum?
- * What if we took the average of the zig-zag steps?

» Momentum Methods: Motivation

- * Oscillations or “chattering” are common when function is non-smooth. For example:

✓ * $f(x) = |x|$, starting value $x_0 = 1$, constant step size $\alpha = 0.1$



- * Function decreases until it reaches vicinity of minimum, then oscillates forever - never converges to minimum.
- * What if we averaged the oscillations and used that value for x ?

» Weighted Sum/Running Average

* $z_{t+1} = \beta z_t + u_t$ with $z_0 = 0$, $0 < \beta < 1$

* Expanding this out:

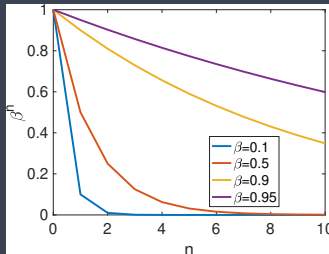
$$\rightarrow z_1 = u_0$$

$$\rightarrow z_2 = \beta z_1 + u_1 = \beta u_0 + u_1$$

$$\rightarrow z_3 = \beta z_2 + u_2 = \beta^2 u_0 + \beta u_1 + u_2$$

so z_t is weighted sum of $u_0, u_1, \dots, u_{t-1}$. E.g. with $\beta = 0.9$ then

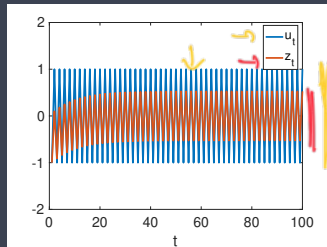
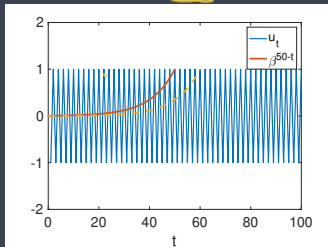
$\rightarrow \beta = 0.9, \beta^2 = 0.81, \beta^3 = 0.729$ etc



» Weighted Sum/Running Average

* Example: $\beta = 0.9$, $u_t = (-1)^t$ i.e. $u_0 = 0, u_1 = 1, u_2 = +1, u_3 = -1, \dots$

* $z_{t+1} = \beta z_t + u_t, z_0 = 0$:

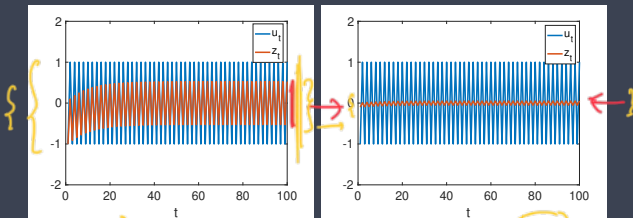


* z_t tends to smooth out (or average) the zig-zags

» Weighted Sum/Running Average

$$(-1)^0 = 1 \quad (-1)^1 = -1$$

* Example: $\beta = 0.9$, $u_t = (-1)^t$ i.e. $u_0 = 0$, $u_1 = 1$, $u_2 = +1$, $u_3 = -1, \dots$



$$z_{t+1} = \beta z_t + u_t$$

$$z_{t+1} = \beta z_t + (1 - \beta) u_t$$

* $1 + \beta + \beta^2 + \dots = 1/(1 - \beta)$, so often scale z_t by $1 - \beta$ i.e. use

$$z_{t+1} = \beta z_t + (1 - \beta) u_t$$

* Another way to think about this: suppose $u_t = \text{constant } u$ and $z_t = \text{constant } z$.

* $z = \beta z + (1 - \beta) u$ rearranges to $(1 - \beta) z = (1 - \beta) u$ i.e. $z = u$.

* $z = \beta z + u$ rearranges to $(1 - \beta) z = u$ i.e. $z = u/(1 - \beta)$.

» Polyak Momentum/Heavy Ball

$x_0 = x_0; z_0 = 0; t = 0$

for k in range(num_iters):

→ $z_{t+1} = \beta z_t + \alpha \nabla f(x_t)$ *step size.*

→ $x_{t+1} = x_t - z_{t+1}$ *new pt*

$t = t + 1$ //

* Step z_t at iteration t is weighted sum of past gradients $\alpha \nabla f$:

* → $z_{t+1} = \underbrace{\beta^{t-1}}_{\text{weight}} \alpha \nabla f(x_0) + \underbrace{\beta^{t-2}}_{\text{weight}} \alpha \nabla f(x_1) + \dots + \underbrace{\alpha \nabla f(x_t)}_{\text{weight}}$

(Handwritten annotations: 1, 2, t are circled; arrows point from 1, 2, t to the corresponding terms in the sum)

* Need to manually select β and α .

* A typical choice for β seems to be 0.9, but there is not a universal value and a poor choice can make performance worse than using constant step size

→ * When $\beta = 0$ recover constant step size strategy

* Since $1 + \beta + \beta^2 + \dots = 1/(1 - \beta)$, we're effectively using scaling baseline step size $\alpha/(1 - \beta)$ i.e. for $\beta = 0.9$ using 10α

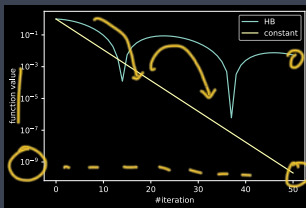
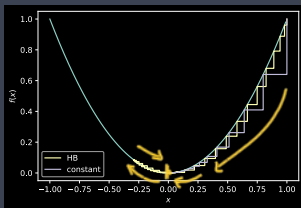
» Too Much Momentum

* Example: $f = x^2$, starting point $x = 1$

* HB $\beta = 0.9$, $\alpha_0 = 0.01$, constant $\alpha_0 = 0.1$ (so HB α_0 is $1 - \beta$ times constant α_0)

$$\alpha_0 = (1 - \beta) \alpha_{\text{const}_0}$$

①



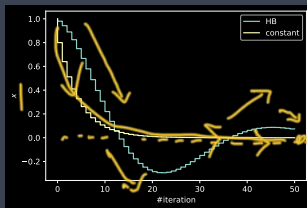
10^{-3}

②

10^{-9}

* Adding momentum seems to have introduced new oscillations!
What's going on here ...

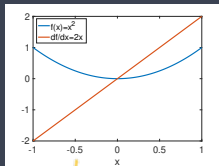
* Hint: see that x overshoots (goes past) the minimum at $x = 0$



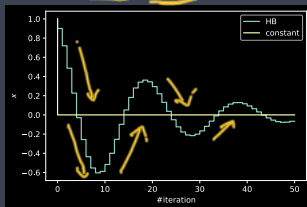
③

» Too Much Momentum

* Example: $f(x) = x^2$, $\frac{df}{dx}(x) = 2x$



→ * Step at iteration t is $z_{t+1} = \underbrace{\beta^{t-1}}_{\text{momentum}} \underbrace{\alpha \frac{df}{dx}(x_0)}_{\text{step size}} + \underbrace{\beta^{t-2}}_{\text{momentum}} \underbrace{\alpha \frac{df}{dx}(x_1)}_{\text{step size}} + \cdots + \underbrace{\alpha \frac{df}{dx}(x_t)}_{\text{step size}}$



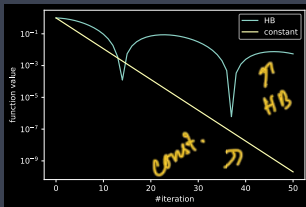
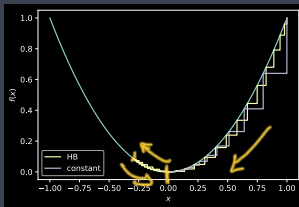
- * $x \approx 0$ around #iteration 12, but HB step size is high, so shoot right past the minimum.
- * x then becomes negative, and so does df/dx . HB step size starts to decrease → but takes a while for step to become negative due to momentum/averaging, so x overshoots
- * $x \approx 0$ around #iteration 38, HB step size again big, so overshoot

» Examples

* Example: $f = x^2$, starting point $x = 1$

→ * HB $\beta = 0.9$, $\alpha = 0.01$, constant $\alpha_0 = 0.1$

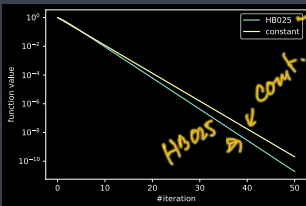
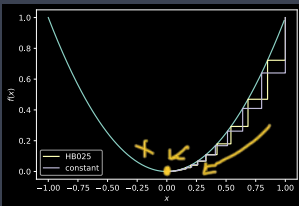
①



②

→ * HB $\beta = 0.25$, $\alpha = 0.075$, constant $\alpha_0 = 0.1$

③



④

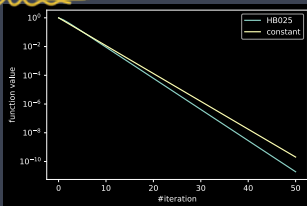
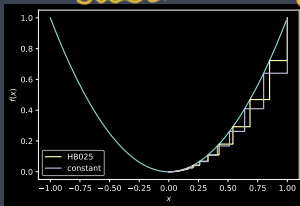
* Decreasing β (reducing momentum i.e. weight given to past gradients) removes the oscillations.

» Examples

* Example: $f = x^2$, starting point $x = 1$

→ * HB $\beta = 0.25$, $\alpha = 0.075$, constant $\alpha_0 = 0.1$ ✓

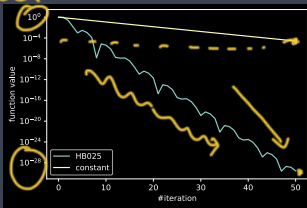
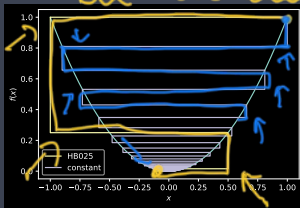
①



②

→ * HB $\beta = 0.25$, $\alpha = 1$, constant $\alpha_0 = 0.95$ ✓

③



10^{-4}
④
 10^{-28}

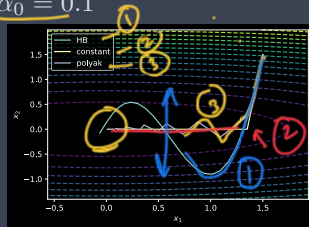
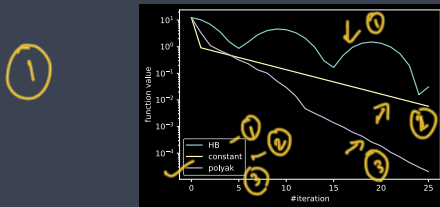
* For constant step size, $\alpha_0 > 1$ gives divergent solution, and very oscillatory solutions for $\alpha_0 > 0.9$.

* With HB can use larger step size without destabilising solution → faster convergence

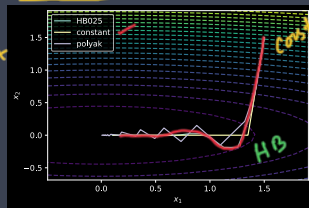
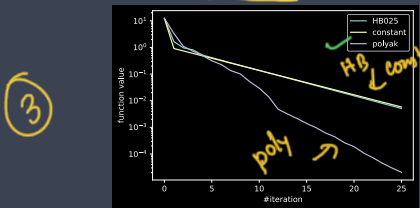
» Examples

* Quadratic:

* HB $\beta = 0.9$, $\alpha = 0.01$, constant $\alpha_0 = 0.1$



* HB $\beta = 0.25$, $\alpha = 0.075$, constant $\alpha_0 = 0.1$



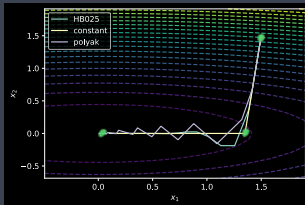
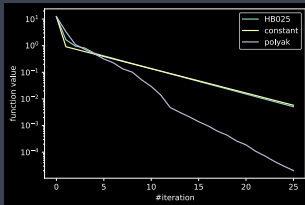
* Again, HB overshoots minimum when $\beta = 0.9$ and increases oscillations. Using less momentum $\beta = 0.25$ removes oscillations $\rightarrow$ performance no better than constant step size strategy but with HB can increase step size

» Examples

* Quadratic:

* HB $\beta = 0.25$, $\alpha = 0.075$, constant $\alpha_0 = 0.1$

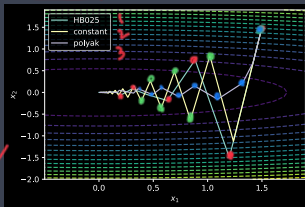
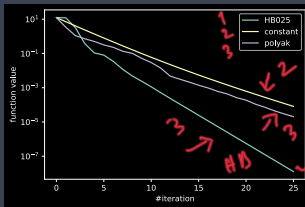
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* Increase step size: HB $\beta = 0.25$, $\alpha = 0.2$, constant $\alpha_0 = 0.175$

③



④

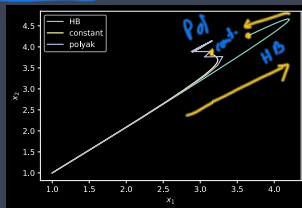
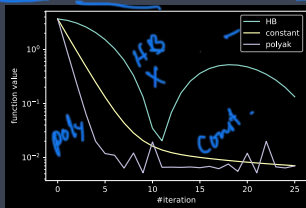
* Increasing constant step size α_0 to 0.2 causes solution to diverge

» Examples

→ * Quadratic loss: ↓

* HB $\beta = 0.9$, $\alpha = 0.05$, constant $\alpha_0 = 0.5$

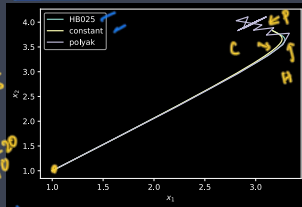
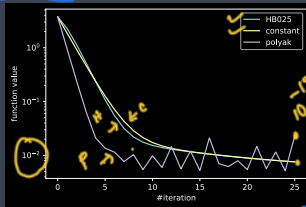
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→ * HB $\beta = 0.25$, $\alpha = 0.375$, constant $\alpha_0 = 0.5$

③



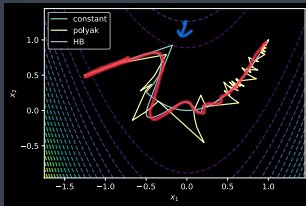
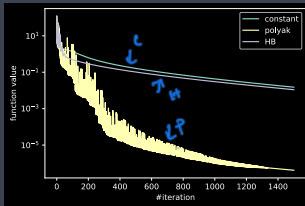
④

* Similarly to last example: $\beta = 0.9$ increases oscillations, $\beta = 0.25$ offers little benefit but wit HB can increase step size

» Examples

- * Rosenbrock function:
- * HB $\beta = 0.9$, $\alpha = 0.0002$, constant $\alpha_0 = 0.002$

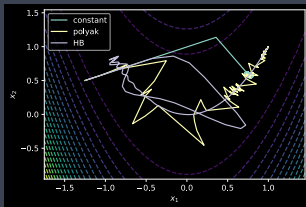
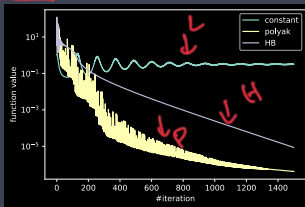
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②

- * Finally an example where $\beta = 0.9$ doesn't increase oscillations.
- * Performance of HB is similar to constant step size when HB $\alpha = 0.0002$ is $(1 - \beta)$ times constant $\alpha_0 = 0.002$
- * HB allows use of larger α without destabilising solution. HB $\beta = 0.9$, $\alpha = 0.0003$, constant $\alpha_0 = 0.008$

③



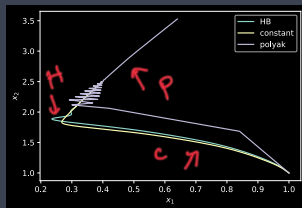
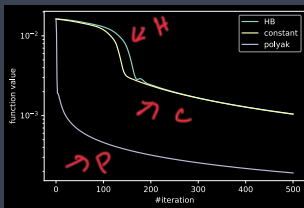
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» Examples

* Toy neural net loss:

* HB $\beta = 0.9$, $\alpha = 0.075$, constant $\alpha_0 = 0.75$ ✓

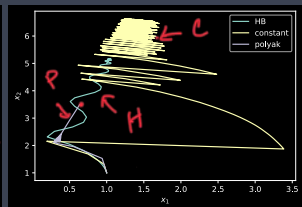
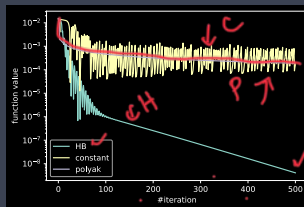
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* Again HB allows larger α than constant step strategy e.g. HB $\beta = 0.9$, $\alpha = 7.5$, constant $\alpha_0 = 75$ ✓

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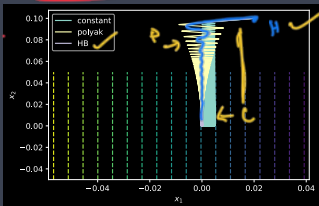
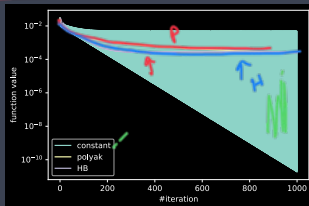


④

» Examples

- * Non-smooth function $f(x) = |x_1| + x_2^2$
- * HB $\beta = 0.9$, $\alpha = 0.0005$, constant $\alpha_0 = 0.005$

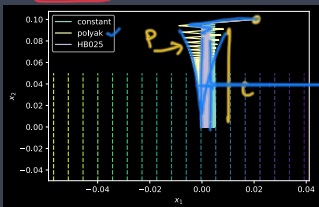
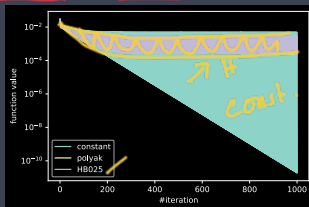
1



2

- * HB $\beta = 0.25$, $\alpha = 0.004$, constant $\alpha_0 = 0.005$

3



4

- * Higher momentum ($\beta = 0.9$) reduces oscillations, as expected
- * In this example can't increase HB α further without destabilising, but already faster than constant step size strategy

» Summary

- * Use of momentum can reduce oscillations/chattering
- * But need to manually tune momentum parameter β , a poor choice can *increase* oscillations/chattering
- * Also need to manually tune α
- * HB can allow use of a larger step size α without causing solution to diverge $\rightarrow$ faster convergence than with constant step size strategy